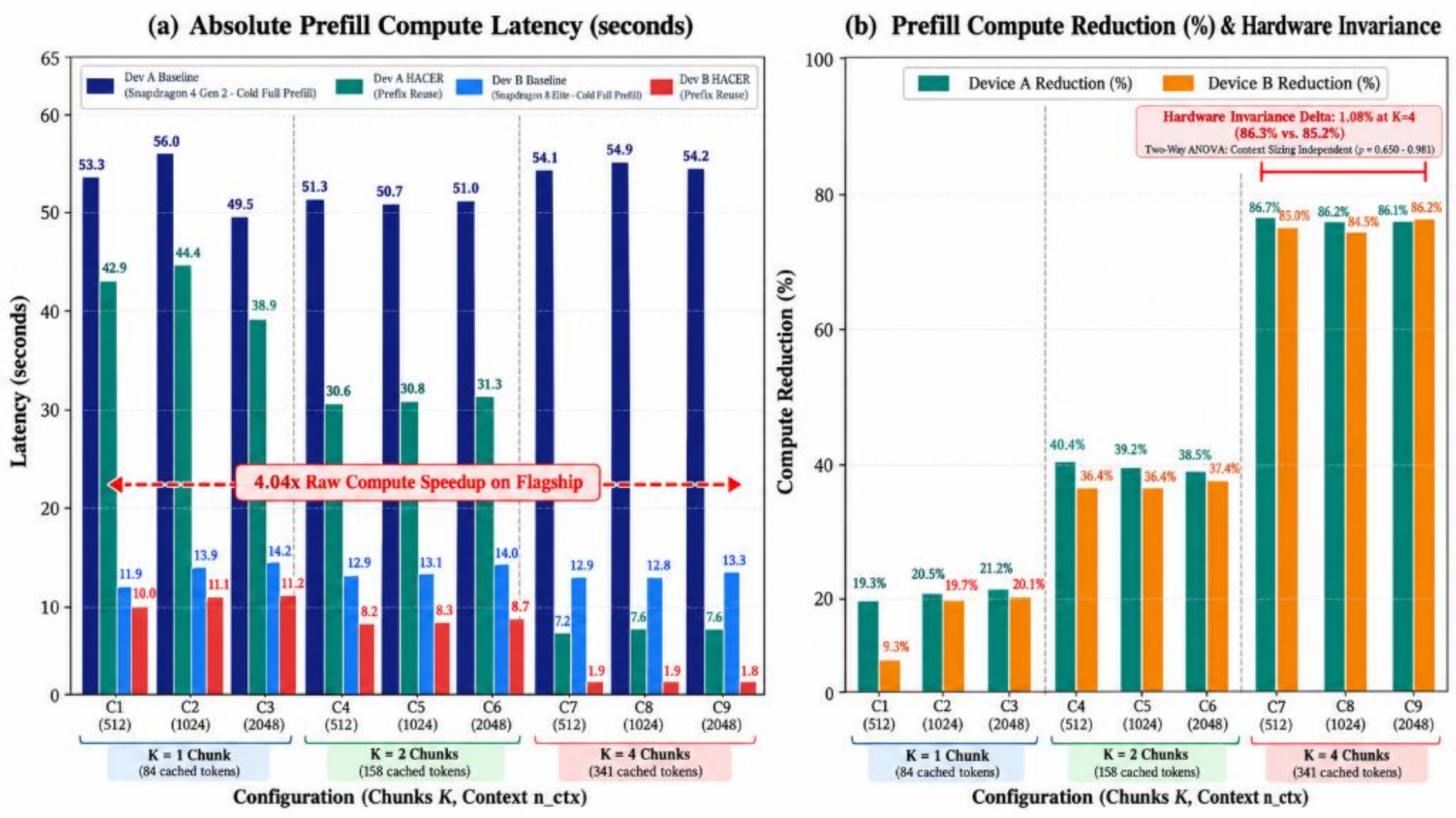

(a) Absolute Prefill Compute Latency (seconds)
Dev A Baseline (Snapdragon 4 Gen 2 - Cold Full Prefill)
Dev A HACER (Prefix Reuse)
Dev B Baseline (Snapdragon 8 Elite - Cold Full Prefill)
Dev B HACER (Prefix Reuse)
Latency (seconds)
4.04x Raw Compute Speedup on Flagship
C1 (512)
C2 (1024)
C3 (2048)
C4 (512)
C5 (1024)
C6 (2048)
C7 (512)
C8 (1024)
C9 (2048)
K = 1 Chunk (84 cached tokens)
K = 2 Chunks (158 cached tokens)
K = 4 Chunks (341 cached tokens)
Configuration (Chunks K, Context n_ctx)
(b) Prefill Compute Reduction (%) & Hardware Invariance
Device A Reduction (%)
Device B Reduction (%)
Hardware Invariance Delta: 1.08% at K=4 (86.3% vs. 85.2%)
Two-Way ANOVA: Context Sizing Independent (p = 0.650 - 0.981)
Compute Reduction (%)
Configuration (Chunks K, Context n_ctx)